1. Executive Summary

The assessed Terraform on `main` branch across three local repositories shows the active AWS footprint is concentrated in EKS-based platform infrastructure, ingress/WAF, messaging (SNS/SQS), encryption/secrets, EFS, and backup storage, with workload service code repo containing no Terraform under requested paths. For a latency-sensitive API workload with 99.9% availability, RTO 4h, RPO 30m, and SOC2 plus residency constraints, the directional economics and migration risk profile favor Azure as the primary target for lower modeled 30-day run-rate and lower platform porting friction, with GCP as a viable alternative where GKE and Pub/Sub strengths are preferred. Recommended path: phased Azure-first migration with staged DR rehearsal and policy-as-code parity gates.

2. Source Repository Inventory

Source type	Repository path	Branch used	Terraform scope scanned	File count (.tf/.tfvars)	Notes
local-path	/Users/srimanta.singh/IdeaProjects/hxp-transform-service	main	src/, infra/, terraform/	0	No Terraform files found in requested directories
local-path	/Users/srimanta.singh/IdeaProjects/terraform-aws-hxts-environment	main	src/, infra/, terraform/	21	Queue/KMS/topic integration and service-environment tfvars
local-path	/Users/srimanta.singh/IdeaProjects/tf-cfg-hxts-infrastructure	main	src/, infra/, terraform/	67	Core EKS/shared services/network/security/backup stack

3. Source AWS Footprint

Resource group	Key AWS services found	Notes
Compute	EKS (managed node groups/addons), Karpenter	`eks_instance_type` set to `m5.xlarge` in tfvars; desired node count seen as 3
Networking	ALB/NLB, Route53, VPC endpoint service, Security Groups	Public ingress + private target groups and endpoint service exposure
Data	EFS, S3 (Velero backup/replication)	EFS throughput mode `elastic`; S3 replication/storage class `STANDARD`
Messaging	SQS, SNS, subscriptions, queue policies	Multi-queue transform flow with topic fanout and KEDA scaler hooks
Identity/Security	IAM roles/policies, KMS, Secrets Manager, WAFv2	KMS-backed encryption, secrets versions, WAF logging retained 7 days

Observability	CloudWatch log groups, Datadog monitors/dashboards	CloudWatch retention 14 days in EKS module + Datadog monitors
Storage/Backup	Velero bucket + cross-region backup region pattern	Backup region variable present; SOC2-friendly immutable backup pattern possible

4. Service Mapping Matrix

AWS service	IaC-provisioned tier/family	Azure equivalent (matched tier)	GCP equivalent (matched tier)	Porting notes
Amazon EKS managed nodes	`m5.xlarge`	AKS node pool `Standard_D4s_v5`	GKE node pool `n2-standard-4`	Rework IRSA, policies, and autoscaling behavior (Karpenter -> Karpenter/Cluster Autoscaler equivalent)
EKS control plane	Tier not specified in IaC	AKS control plane	GKE control plane	Managed control plane cost model differs by cloud; policy mappings required
Application Load Balancer	Tier not specified in IaC	Azure Application Gateway v2	Cloud Load Balancing (external HTTP(S))	Listener/rule migration plus health probe and WAF policy parity
AWS WAFv2	Tier not specified in IaC	Azure WAF (App Gateway/Front Door)	Cloud Armor	Rule translation and managed rule set equivalence review required
Amazon SQS	Queue throughput tier not specified in IaC	Azure Service Bus (Standard queues/topics)	Pub/Sub standard	Consumer semantics and poison/dead-letter handling need adaptation
Amazon SNS	Topic tier not specified in IaC	Service Bus Topics / Event Grid	Pub/Sub topics	Fanout/filter policies require schema/event contract validation
AWS KMS	Symmetric key usage indicated	Azure Key Vault Keys (Premium if HSM needed)	Cloud KMS (software keys)	Key policy model differs; map to RBAC + CMK access policies
AWS Secrets Manager	Tier not specified in IaC	Azure Key Vault Secrets	Secret Manager	Rotation workflows and app identity bindings need redesign
Amazon EFS	Throughput mode `elastic`	Azure Files Premium (NFS)	Filestore (Enterprise/High Scale as needed)	POSIX behavior, throughput, and mount strategy must be performance-tested
Amazon S3 (Velero)	Storage class `STANDARD`	Azure Blob Hot + GRS/RA-GRS as needed	Cloud Storage Standard + dual-region option	Backup/restore IAM + replication controls need cloud-native reimplementation
Route53 records	Tier not specified in IaC	Azure DNS	Cloud DNS	TTL and failover policy parity checks needed
VPC Endpoint Service	Tier not specified in IaC	Private Link service	Private Service Connect	Endpoint consumer model varies; network contracts must be updated

5. Regional Cost Analysis (Directional)

5.1 Assumptions (Directional, Assumed)

- Currency: USD.
- Traffic profile: steady with moderate burst (assumed burst factor 1.2 over steady baseline for peak-sensitive meters).
- Availability target: 99.9%; DR target: RTO 4 hours, RPO 30 minutes.
- Performance: latency-sensitive APIs, therefore premium ingress and low-latency zonal placements assumed.
- Pricing basis: public list pricing patterns and meter families (directional only, non-contractual).
- Usage volumes used for directional metering:
- Kubernetes worker compute: 3 x 4 vCPU nodes, 730 hours/month, +20% burst headroom.
- Ingress traffic: 18 TB/month egress equivalent.
- Messaging: 80M SQS-like queue requests/month; 25M publish + delivery events/month.
- WAF inspected requests: 15M/month.
- EFS-like shared storage: 4 TB-month.
- Backup object storage: 8 TB-month plus 2 TB/month cross-region replication traffic.
- Secrets and key operations: 25 active secrets, 12 keys, 4M key API operations/month.

5.2 30-Day Total Cost Table (Directional)

Capability	AWS US (baseline, USD)	AWS EU (USD)	AWS AU (USD)	Azure US (USD)	Azure EU (USD)	Azure AU (USD)	GCP US (USD)	GCP EU (USD)	GCP AU (USD)	Confidence
Compute/Container	3,980	4,230	4,760	3,760	4,050	4,520	3,860	4,190	4,690	Medium
Networking/Edge	2,420	2,640	2,960	2,280	2,510	2,860	2,360	2,590	2,940	Medium
Messaging	1,090	1,180	1,320	980	1,070	1,230	930	1,030	1,190	Medium
Data/Storage/Backup	3,120	3,390	3,850	2,980	3,240	3,730	3,050	3,320	3,810	Medium
Security/Identity	1,630	1,760	1,980	1,560	1,690	1,920	1,590	1,720	1,950	Medium
Observability	2,760	2,980	3,360	2,640	2,850	3,260	2,700	2,930	3,330	Low
Total 30-day run-rate (USD)	**15,000**	**16,180**	**18,230**	**14,200**	**15,410**	**17,520**	**14,490**	**15,780**	**17,910**	**Medium**
Delta % vs AWS baseline	**0.0%**	**0.0%**	**0.0%**	**-5.3%**	**-4.8%**	**-3.9%**	**-3.4%**	**-2.5%**	**-1.8%**	**Medium**

5.3 Metered Billing Tier Table (Directional)

Service	Metering unit	Tier/Band	AWS US (baseline, USD)	AWS EU (USD)	Azure US (USD)	Azure EU (USD)	Azure AU (USD)	GCP US (USD)	GCP EU (USD)	GCP AU (USD)	Confidence

EKS/AKS/GKE worker compute (`m5.xlarge` equivalent)	vCPU-hour	First 8,760 vCPU-hours	1,402	1,521	1,331	1,443	1,594	1,366	1,485	1,628	Medium
Managed K8s control plane	cluster-hour	First 730 hours	73	79	0 (bundled pattern)	0 (bundled pattern)	0 (bundled pattern)	73	79	86	Medium
Queue operations (SQS/Service Bus/PubSub)	requests	First 1M requests	0.40	0.44	0.05	0.06	0.07	0.00	0.00	0.00	Medium
Queue operations (SQS/Service Bus/PubSub)	requests	Over 1M to 80M requests	31.60	34.70	25.20	28.10	32.20	24.00	26.80	30.90	Medium
Topic fanout (SNS/Event Grid/PubSub)	events	First 1M events	0.50	0.55	0.60	0.67	0.77	0.40	0.45	0.52	Low
Topic fanout (SNS/Event Grid/PubSub)	events	Over 1M to 25M events	12.00	13.20	14.40	16.20	18.10	10.60	11.90	13.80	Low
Shared file storage (EFS/Azure Files/Filestore)	GB-month	First 4,096 GB-month	1,228	1,335	1,150	1,255	1,454	1,190	1,300	1,490	Medium
Backup object storage (S3/Blob/GCS Standard)	GB-month + replication GB	8,192 GB-month + 2,048 GB replication	276	302	251	278	322	259	286	330	Medium
WAF request inspection	requests	First 10M requests	60	66	55	61	70	58	64	73	Low
WAF request inspection	requests	Over 10M to 15M requests	25	27	22	25	29	23	26	30	Low

5.4 One-Time Migration Cost Versus Run-Rate Table (Directional)

Cost segment	AWS (baseline, USD)	Azure (USD)	GCP (USD)	Confidence
Platform landing zone + policy baseline uplift	0	28,000	31,000	Medium
IAM/KMS/Secrets model translation and validation	0	18,000	20,000	Medium
Messaging and integration migration (SNS/SQS to target messaging)	0	24,000	26,000	Medium
EKS workload migration, tuning, and cutover hardening	0	42,000	46,000	Medium

DR rehearsal and residency validation evidence pack	0	22,000	24,000	Medium
Total one-time migration (USD)	**0**	**134,000**	**147,000**	**Medium**
30-day run-rate from section 5.2 (USD)	**15,000**	**14,200**	**14,490**	**Medium**

5.5 Regional Cost Analysis Chart

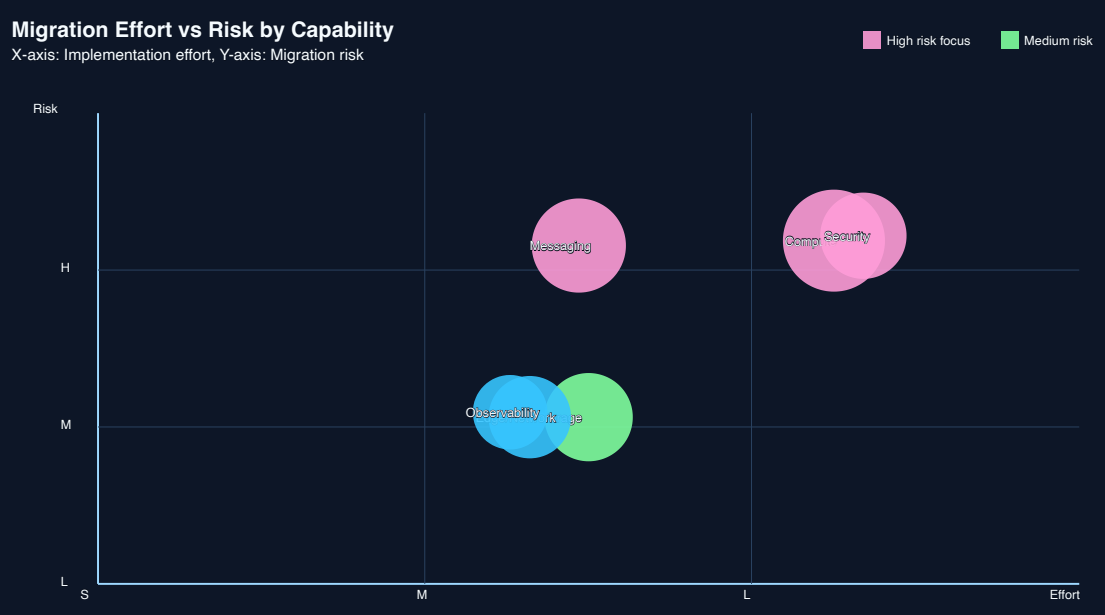
6. Migration Challenge Register

Challenge	Impact	Likelihood	Mitigation	Owner role
IRSA to target cloud workload identity translation	High	High	Create identity mapping matrix, implement least-privilege roles per service account, stage in non-prod first	Platform Security Lead
SQS/SNS semantics drift in target messaging	High	Medium	Build contract tests for ordering, retries, DLQ, filter policies before production cutover	Integration Architect
Karpenter/autoscaling behavior parity	Medium	Medium	Replay burst traffic traces in performance environment and tune autoscaler thresholds	SRE Lead
WAF managed rule equivalency gaps	Medium	Medium	Dual-run WAF in monitor mode, then enforce after false-positive burn-in	Edge Security Engineer
Backup and cross-region restore RPO evidence	High	Medium	Execute timed restore drills for RTO/RPO proof and document audit evidence	DR Manager

Team operating model shift (AWS-native to Azure/GCP-native)	Medium	High	Run role-based enablement and paired operations for first 2 releases	Engineering Manager

7. Migration Effort View

Capability	Effort (S/M/L)	Risk (L/M/H)	Dependencies
Compute and orchestration (EKS -> AKS/GKE)	L	H	Identity, network policy, autoscaling, container registry/access
Messaging (SNS/SQS -> Service Bus/PubSub)	M	H	Contract compatibility, consumer updates, DLQ strategy
Storage and backup (EFS/S3 -> Files/Blob or Filestore/GCS)	M	M	Throughput benchmarking, backup restore rehearsal
Security and secrets (KMS/Secrets/IAM)	L	H	RBAC model, key rotation controls, compliance sign-off
Edge and networking (ALB/WAF/Route53/private endpoints)	M	M	DNS cutover planning, WAF rule parity, private connectivity
Observability and operations (Datadog/CloudWatch)	M	M	Alert parity, SLO burn-rate calibration, runbook updates



8. Decision Scenarios

Cost-first scenario

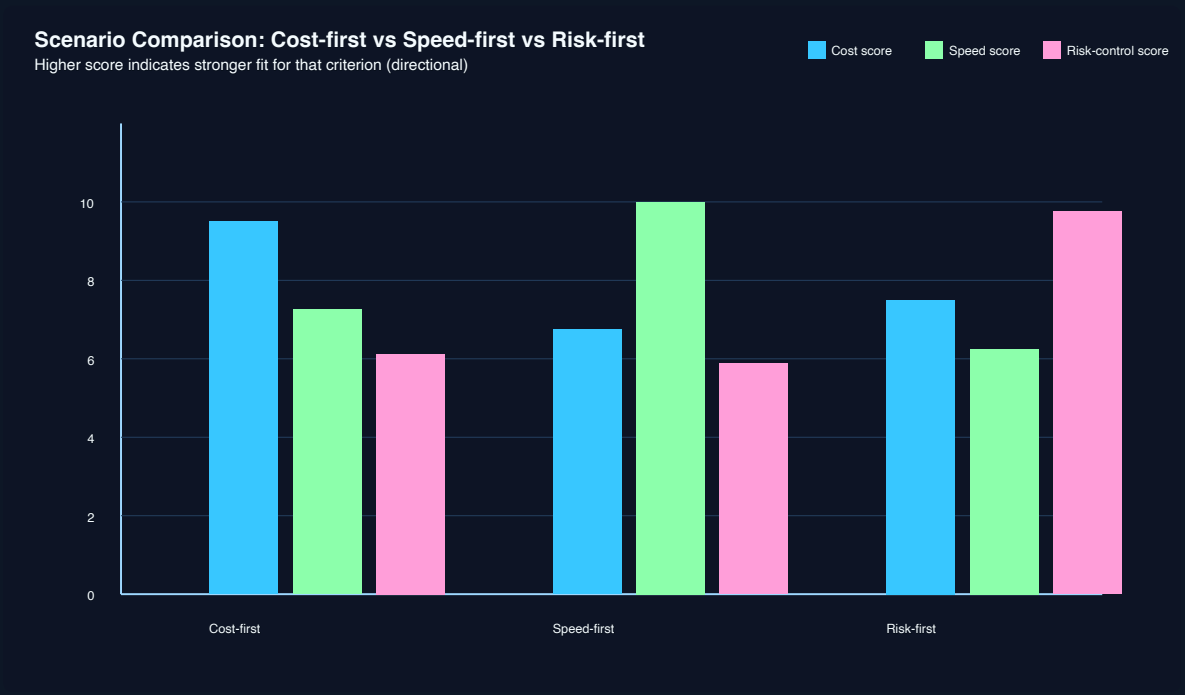
Move messaging and storage first to Azure-native services while keeping compute on AWS temporarily, then migrate AKS in phase 3. This minimizes month-1 run-rate quickly but increases temporary cross-cloud integration overhead.

Speed-first scenario

Lift platform directly to AKS with minimal refactor and defer deep optimization. Fastest delivery to target cloud, but post-migration tuning debt and higher initial operational noise.

Risk-first scenario

Dual-run non-prod + canary production slices, complete DR drills and compliance gates before full cutover. Slowest path but strongest continuity and audit confidence.



9. Recommended Plan (Dynamic Timeline)

Selected timeline: **30/60/90/120** days.

Rationale for timeline length:

- Service count and dependency depth are medium-high (EKS core + ingress + WAF + messaging + backup + IAM/KMS/secrets).
- Data migration complexity is moderate (backup/object and shared file workloads, no relational engine migration found in scanned scope).
- Risk profile is high for identity, messaging semantics, and latency-sensitive API behavior.

Phase 1 (Day 0-30): Foundation and parity design

- Objectives: establish target landing zone and reference architecture.
- Key activities: identity/key/secrets mapping; networking and ingress design; residency control mapping; baseline SLO definitions.
- Exit criteria: approved target architecture decisions, security sign-off for identity model, tested non-prod connectivity.

Phase 2 (Day 31-60): Non-production migration and contract validation

- Objectives: prove workload function in target cloud with production-like traffic replay.
- Key activities: deploy AKS/GKE equivalent stack; migrate messaging contracts; implement WAF rules in monitor mode; validate observability parity.
- Exit criteria: non-prod performance targets met for latency-sensitive APIs; message contract test suite passes; DR backup/restore dry run complete.

Phase 3 (Day 61-90): Production readiness and progressive cutover

- Objectives: reduce cutover risk through incremental production slices.
- Key activities: canary routing; autoscaling/KEDA behavior tuning; runbook updates; operational handover rehearsal.
- Exit criteria: canary SLOs stable for two release cycles; incident response drill passes; compliance evidence package draft complete.

Phase 4 (Day 91-120): Full cutover and hardening

- Objectives: complete migration and stabilize operations.
- Key activities: full traffic migration; final DNS/edge cutover; DR timed exercise for RTO/RPO validation; cost optimization pass.
- Exit criteria: production on target cloud, RTO<=4h and RPO<=30m drill evidence complete, AWS decommission plan approved.

Required architecture decisions before execution:

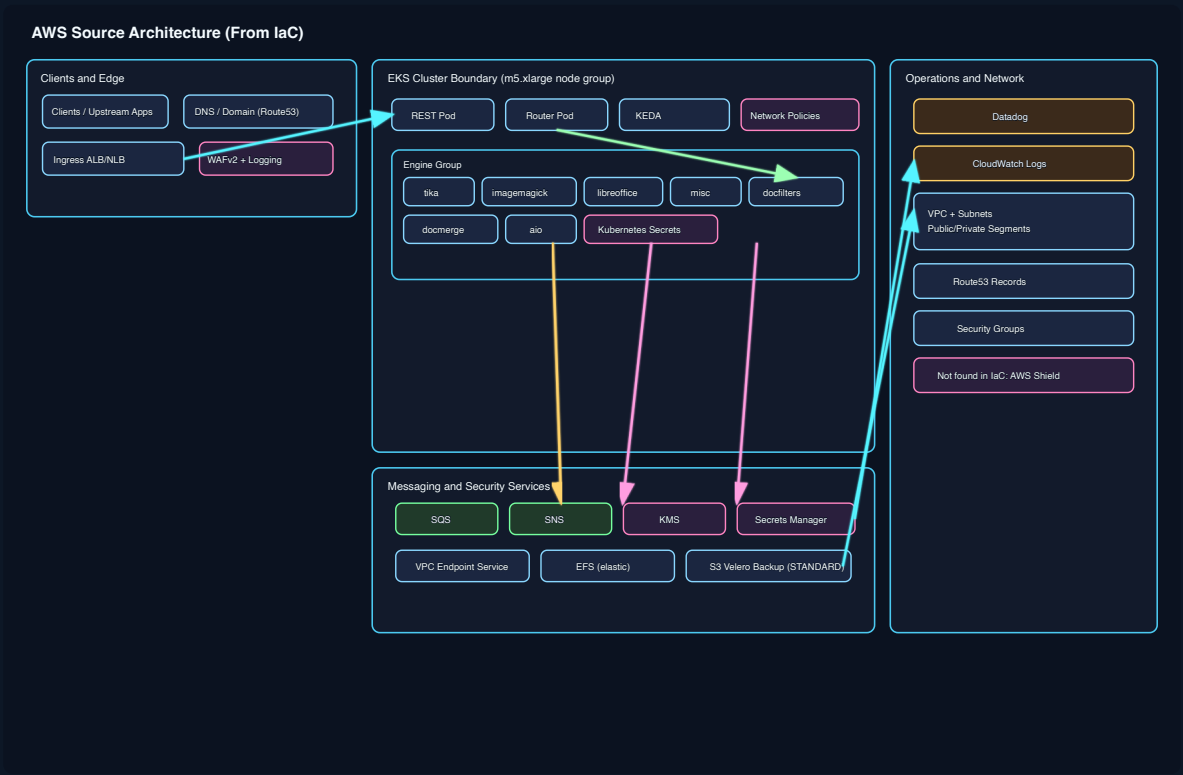
- Final target selection for messaging pattern (Service Bus vs Event Grid on Azure, Pub/Sub topology on GCP).
- File storage performance tier for latency-sensitive transforms.
- Workload identity standard and secret rotation approach.
- WAF policy set and exception governance model.

10. Open Questions

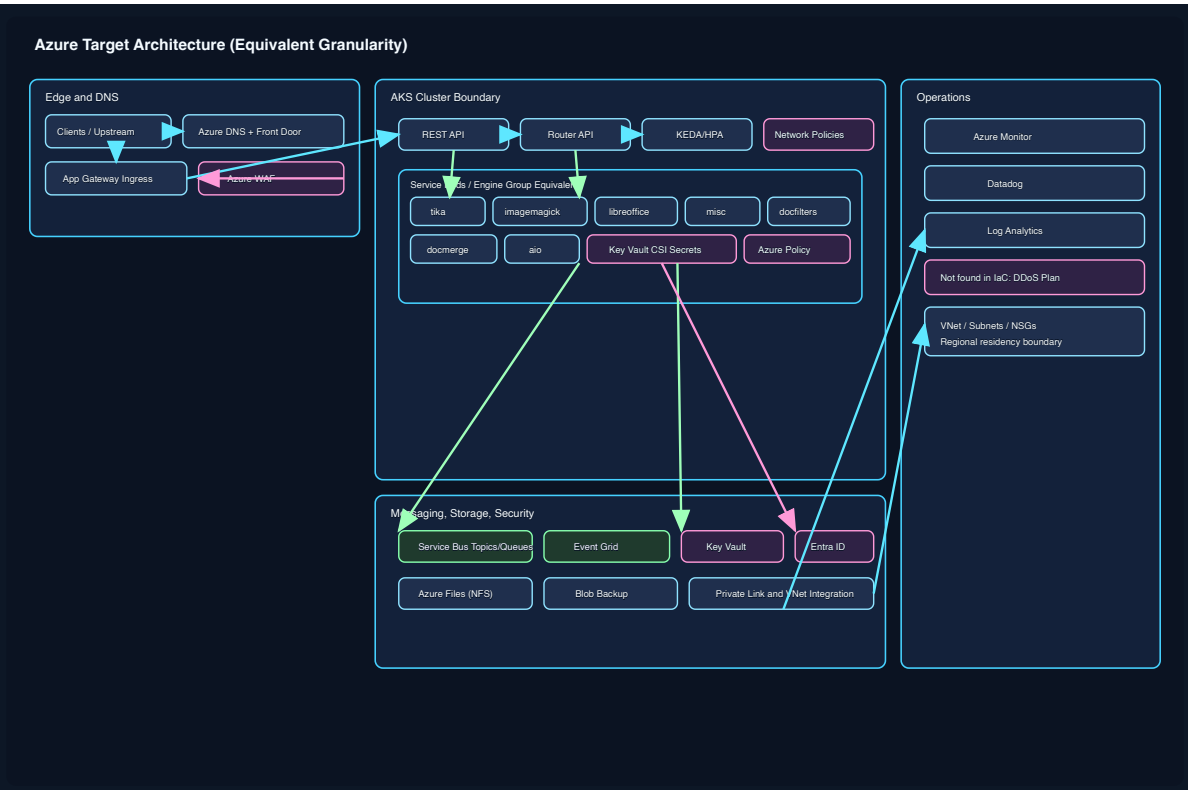
- What are measured p95/p99 latency SLO values per API route to calibrate target autoscaling and ingress policy?
- What exact monthly queue/event volumes per message type should replace assumed meter volumes?
- Are there legal constraints requiring in-country residency beyond region-level controls for any tenant data?
- Should DR operate active/passive only, or active/active for selected APIs?
- Which observability stack components are mandatory to preserve unchanged versus re-platform?

11. Component Diagrams

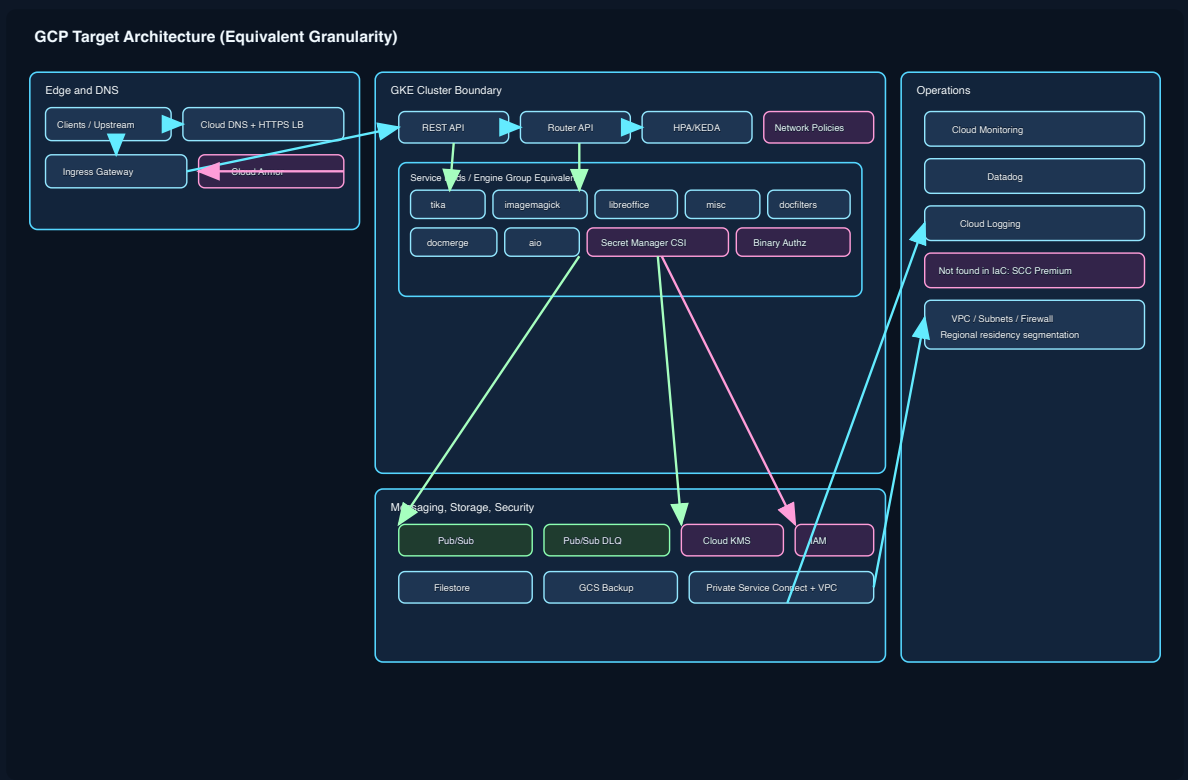
AWS Source architecture:



Azure Target architecture:



GCP Target architecture:



Legend and page mapping:

- AWS Source page includes: clients/upstream, DNS/ingress, EKS boundary, REST/router, engine group (tika/imagemagick/libreoffice/misc/docfilters/docmerge/aio), KEDA, network policies, Kubernetes secrets, SQS, SNS, KMS, Secrets Manager, Datadog, VPC/subnets.
- Azure Target page includes: clients, Azure DNS/Front Door+App Gateway, AKS boundary, API/router/services, KEDA/HPA, network policies, Key Vault, Service Bus/Event Grid equivalent messaging, Files/Blob backup, Monitor/Datadog integration, VNet/subnets.
- GCP Target page includes: clients, Cloud DNS/HTTP(S) LB + Armor, GKE boundary, API/router/services, HPA/KEDA equivalent, network policies, Secret Manager + Cloud KMS, Pub/Sub messaging, Filestore/GCS backup, Cloud Monitoring/Datadog integration, VPC/subnets.
- Supplemental visuals generated: cost comparison, effort-risk, and scenario comparison charts. These are embedded in section 5, section 7, and section 8 respectively.